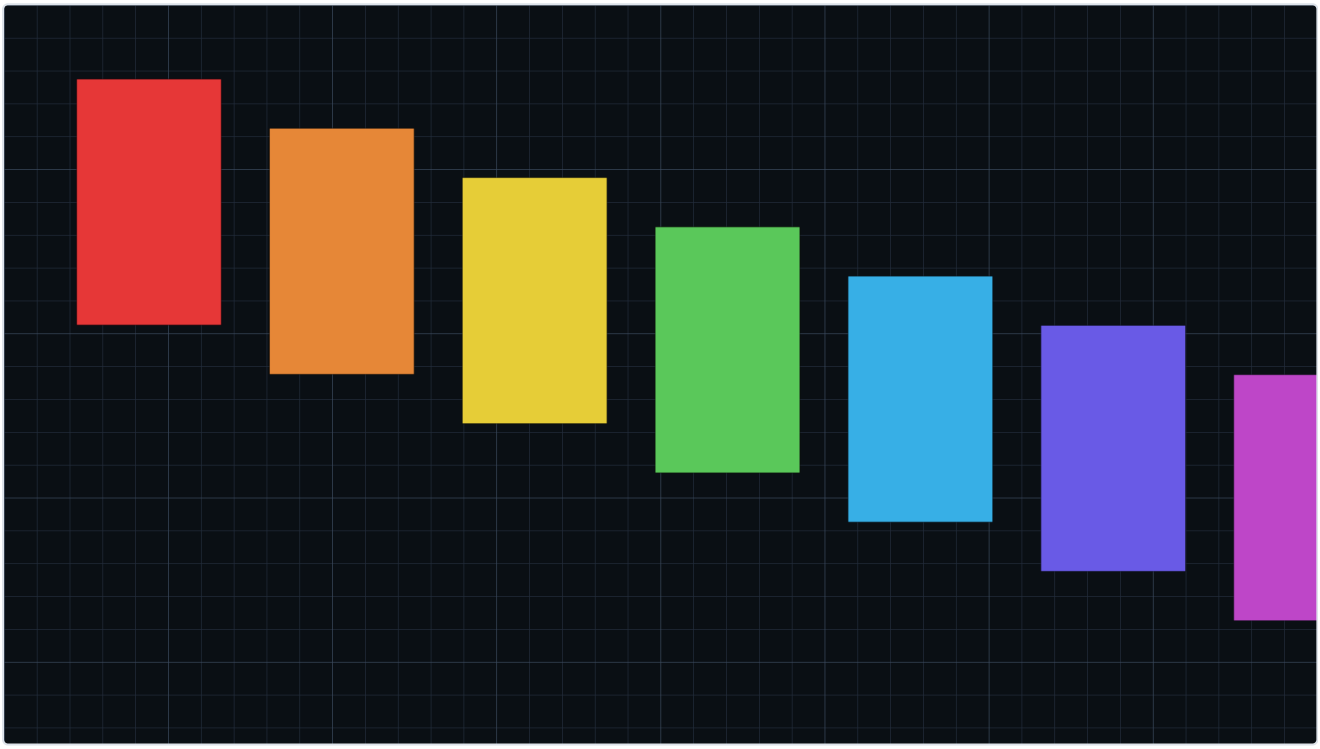
# amber user guide

amber makes **legacy Flash content playable in Resolume**, either by converting `.swf` and `.flv` files to DXV or Hap, or by playing a `.swf` live on a layer with its FFGL plugin.

Resolume won't open `.swf` or `.flv`, so a decade and a half of VJ loops, web animation and motion tests is sitting in formats nothing on the machine will play.



*The repo's generated showcase movie (a stepped diagonal of vector rectangles), rendered by the plugin class through its offline harness with the stage transparent, then composited over a grid, which shows through everywhere the shapes aren't. It's generated demonstration content, nobody else's Flash.*

There are two ways out, and they're good at different things:

- **The converter** turns `.swf` and `.flv` into DXV or Hap (the codecs Resolume decodes on the GPU), keeping the frame rate, transparency and dimensions. Do this for a library.
- **The live plugin** plays a `.swf` on a layer with its timeline and ActionScript actually running. Do this when the movie needs to be interactive, or transparent.

> **Before you rely on this:** I've run the converter end to end against real 2003-era Flash and against real-world FLV in all three of its common codecs, and checked the output is frame-accurate by decoding it again and comparing pixels. 39 tests cover it, and the codec-constraint tests **re-derive** the DXV and Hap dimension rules from the real encoder rather than asserting a table against itself.
>
> **The plugin is confirmed working in Resolume Arena on Apple Silicon.** Its Intel slice has never been run in an Intel Resolume, and the Windows build has never been loaded into a host.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

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## Installing

The converter bundles nothing. It drives your own ffmpeg and your own build of Ruffle's exporter as separate processes, so there's no linking question to answer, and two dependencies to satisfy before it can convert anything. (The live plugin has Ruffle compiled in and needs neither. The licence notices for everything compiled into it are in the repo's `THIRD-PARTY-NOTICES.txt`.)

```
brew install ffmpeg-full
```

**`ffmpeg-full`, not plain `ffmpeg`.** The plain formula has the DXV encoder but **not** the Hap encoder, so transparent content would have nowhere to go.

For `.swf` support you also need Ruffle's exporter, which Ruffle doesn't ship in its releases:

```
git clone --depth 1 https://github.com/ruffle-rs/ruffle ~/Documents/Amber/ruffle-src
cd ~/Documents/Amber/ruffle-src && cargo build --release --package=exporter
```

amber finds it there automatically. `.flv` conversion works without it.

**Then run `amber doctor` before anything else.** It reports which ffmpeg it found, whether that build has Hap, and whether the Ruffle exporter is present, which is the difference between "this file cannot be converted" and "this machine cannot convert that file yet".

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## Converting

```
amber doctor                          # what can this machine do?
amber info  ~/clips/badger.swf        # describe it without converting
amber convert ~/clips -o ~/converted  # a whole directory
```

```
badger.swf → badger.mov  544x400 @ 25fps  904 frames  15.2MB  [dxv]  [resized 550x400 → 544x400 (scale)]
```

The output profile is chosen from what the source needs rather than from a default you have to remember:

| Profile | When | |
|---|---|---|
| `dxv` | opaque content | Resolume's own codec, GPU-decoded |
| `hap_alpha` | the source has alpha | GPU-decoded, alpha byte-exact |
| `hap_q` | on request | 46.8 dB against DXV's 45.0, worst-case error halved, ~80% larger |
| `prores4444` | alpha fallback where Hap is unavailable | CPU-decoded, much larger |

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## Three things about DXV, before you convert a library

**DXV can't carry transparency, and doesn't say so.** ffmpeg's DXV encoder offers just the one format, whose own help text reads "Normal Quality, No Alpha". Hand it an image with alpha and it doesn't warn you, it just returns every pixel opaque. Since VP6-alpha exists for transparent overlay loops in the first place, amber refuses to send alpha content to DXV and routes it to Hap Alpha instead. `--flatten` overrides that if you actually want the alpha gone.

**DXV silently corrupts any width that isn't a multiple of 16.** The encoder returns success, writes a plausible file and records the right dimensions, and the picture inside is sheared diagonally. That would be a corner case, except that **550×400 is the default Flash stage size**, so a naive pipeline mangles a large fraction of all Flash ever made.

amber measures the constraint and works around it. For `.swf` that costs nothing: Flash is vector, so Ruffle rasterises straight to a legal size. For `.flv` the frames are raster and have to be resampled ( `--fit scale` , the default) or padded ( `--fit pad` ).

**DXV blocks up on smooth gradients.** It's DXT1 at a fixed 4:1, so skies and soft falloffs show visible 4×4 blocking. Measured against a lossless reference on gradient-heavy 720p: DXV 45.0 dB with a worst-case error of 44; `hap_q` 46.8 dB with a worst-case error of 19, for about 80% more file. Reach for `--profile hap_q` when the content is gradient-heavy and the disk can take it. **Plain `--profile hap` is pointless**: identical quality to DXV and 27% bigger.

And plan disk space: 28 seconds of 3840×2160 came out at 1.4 GB. That's the codec working as intended, trading size for GPU-decodable playback.

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## The live plugin

Install `Amber.bundle` into `~/Library/Graphics/FreeFrame Plug-Ins` and it appears as a **source** rather than an effect, since it generates a layer instead of processing one.

| Parameter | |
|---|---|
| **Movie** | the `.swf` to play |
| **Run** / **Restart** | transport |
| **Speed** | playback rate |
| **Scaling** | Fit / Fill / Stretch |
| **Smoothing** | filtering on the rasterised frame |
| **Transparent** | on by default |

**Transparent is what makes Flash usable as an overlay.** Ruffle implements Flash's own `wmode=transparent`, so everything the movie doesn't draw comes through clear and the layers underneath show. Turn it off to get the movie's declared stage colour instead.

**Only the plugin can do transparency; the converter can't.** The converter drives Ruffle's `exporter` binary, which has no background option and always renders opaque. So a `.swf` that needs a transparent background has to be played live. That's a gap in the exporter's command line rather than in Ruffle, and the converter tells you rather than quietly handing back an opaque clip.

### Three limits worth knowing before a show

- **There's no audio.** FFGL has no audio path at all, so a live Flash clip is silent no matter what it contains. The converter has the same limit.
- **Ruffle runs inside Resolume's own process.** Every call into it is guarded, which turns most bad content into a black layer rather than a crash, but a guard isn't a process boundary, and content that hard-crashes Ruffle takes Resolume down with it.
- **Only the Apple Silicon slice has been run in a host.** The bundle is universal and a Windows build exists; neither the Intel slice nor the Windows DLL has been loaded into a Resolume.

Live playback is `.swf` only. `.flv` goes through the converter.

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## If it goes wrong

**`.swf` files are refused or skipped.** Ruffle's exporter is missing, and `amber doctor` will say so. `.flv` doesn't need it.

**Transparent content came back opaque.** Either it went to DXV with `--flatten`, or it was a `.swf` put through the converter, which can't preserve alpha at all. Play it live instead.

**The picture is sheared diagonally.** A width that isn't a multiple of 16 reached DXV. amber guards against this, so seeing it means something went round the converter.

**Everything looks blocky in the sky.** DXV's fixed 4:1 compression. Convert that clip again with `--profile hap_q`.

**Resolume disappeared while a Flash clip was playing.** Ruffle hard-crashed inside Resolume's process. That content can't be played live until an out-of-process helper exists, so convert it instead.

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